

Binomial Coefficients in Expansion and Probability

One set of numbers, two jobs: binomial coefficients as the terms of an expansion and as the counts inside a probability

Directions. The number $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ does two things on this sheet, and the point of mixing the problems is that you have to notice which one is wanted.

- **In an expansion:** $(a + b)^n = \sum \binom{n}{k} a^k b^{n-k}$. The coefficient of a term is $\binom{n}{k}$ *times* whatever powers of the constants that term carries — forgetting the second factor is the usual slip.
- **In a count:** $\binom{n}{k}$ is the number of ways to choose k objects from n when order does not matter.
- **In a probability:** the same count appears on top, on the bottom, or both. With equally likely outcomes, use $\frac{\text{favourable}}{\text{total}}$; for n independent trials each succeeding with probability p , use $\binom{n}{k} p^k (1-p)^{n-k}$.

Leave probabilities as exact fractions unless a problem says otherwise. Show the binomial coefficient you used, not just the final number.

1. Evaluate $\binom{7}{3}$ using the factorial formula. Show the cancellation.

Answer: _____

2. Expand and simplify: $(x + 2)^3$

Answer: _____

3. A club has 9 members, 4 of whom are seniors. A committee of 3 members is chosen at random.

(a) How many committees of 3 can be formed altogether?

(b) How many of them consist entirely of seniors?

(c) What is the probability that a randomly chosen committee is all seniors? Give an exact fraction.

Answer: _____

4. Pascal's rule. Evaluate $\binom{9}{3}$, $\binom{9}{4}$ and $\binom{10}{4}$, then state the relationship between the three values and explain where it appears in Pascal's triangle.

Answer: _____

5. Expand and simplify: $(2x - 1)^4$

Every term carries a power of 2 as well as a binomial coefficient.

Answer: _____

6. A fair coin is flipped 6 times.

(a) In how many of the 64 equally likely outcomes are exactly 4 of the flips heads?

(b) What is the probability of getting exactly 4 heads? Give an exact fraction.

Answer: _____

7. Find the coefficient of x^5 in the expansion of $(x + 3)^8$.

The binomial coefficient on its own is not the answer.

Answer: _____

8. A bag holds 5 red marbles and 7 blue marbles. Three marbles are drawn at random, all at once.

(a) In how many ways can 2 red marbles be chosen from the red ones, and 1 blue marble from the blue ones?

(b) How many three-marble draws are possible altogether?

(c) Find the probability that exactly 2 of the three marbles are red. Give an exact fraction in lowest terms.

Answer: _____

9. A basketball player makes 70% of her free throws, and the throws are independent. She takes 5 free throws.

Find the probability that she makes exactly 3 of them. Give an exact fraction.

Answer: _____

10. Find the constant term in the expansion of $\left(2x + \frac{1}{x}\right)^6$.

Write the general term first and decide which value of k makes the power of x equal to zero.

Answer: _____

11. Jo says: “A fair coin is flipped 5 times. The probability of getting exactly 2 heads is $\frac{2}{5}$, because 2 of the 5 flips have to be heads.”

- (a) How many of the 32 equally likely outcomes have exactly 2 heads?
- (b) Find the correct probability as an exact fraction.
- (c) Explain Jo’s mistake in one sentence.

Answer: _____

12. Synthesis challenge. An inspector tests 8 components. Each one is defective, independently of the others, with probability $\frac{1}{4}$.

(a) In the expansion of $(q + p)^8$, where p is the probability a component is defective and $q = 1 - p$, one term gives the probability that exactly 2 components are defective. Write that term, including its binomial coefficient.

(b) Evaluate that term as an exact fraction.

(c) Explain why the binomial coefficients appearing in this expansion are exactly the counts of *which* components are defective.

Answer: _____